Fourth Biweekly Project Progress Report

Project: PPT Outline Intelligent Generation and Content Completion System

Reporting Period (Sprint 4): 2026-05-18 - 2026-05-31

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1. Overall Progress Summary

Sprint 4 is the project's reliability-hardening and experimental-validation sprint, aligned with the Updated Project Charter's Gantt phases "Integration & Testing (05-11 -> 05-25)" and "Experimental Validation & Analysis (05-18 -> 06-01)". With the end-to-end prototype already integrated, this sprint focused on stabilizing the live generation path and running the quantitative/qualitative evaluation tracks. Headline outcomes this sprint:

- **Robust error handling (WBS 3.3).** LLM-call failures are now mapped to clear, user-facing messages (auth / model-not-found / rate-limit / timeout / network / 5xx) without leaking upstream URLs or tokens; a per-task hard-timeout fallback was added to the worker.
- **SSE streaming hardened (WBS 3.3 / 2.4).** Server-Sent Events were hardened with a one-time, 30-second SSE ticket (EventSource cannot carry an Authorization header), plus unified SSE framing, heartbeats, and connection confirmation; the user frontend was refactored onto a dedicated task-stream hook.
- **Retrieval performance (WBS 2.3).** Multi-chapter RAG retrieval was batched (one embedding round-trip per ~10 queries instead of one per query), materially reducing content-completion latency and API call volume.
- **Citation correctness (WBS 2.2).** Citation-style rules in the slide-generation prompts were tightened (default summary; the data tag is forbidden on content without figures) and a citation-style normalization pass was added, reducing mislabeled/over-claimed citations in generated slides.
- **Evaluation tracks advanced (RBS-3).** The automated evaluation system was executed and the multi-model comparison and blind human-assessment tracks were run by the testing members, on top of the stabilized pipeline.

Total hours invested this sprint: approximately 42 hours, distributed evenly across the three reporting members (Sprint 1: 68 hrs; Sprint 2: 64 hrs; Sprint 3: 78 hrs). One team member did not submit a report this period and is excluded. The project remains on track for the firmly scheduled final delivery date of June 14, 2026.

2. Key Achievements This Sprint (RBS-aligned)

Structured Outline Generation (RBS-1):

Multi-model horizontal comparison (Qwen / GLM / DeepSeek) executed on the outline-generation task (generation stability, output quality, response efficiency) for the model-selection recommendation.

Knowledge Retrieval & Content Completion (RBS-2):

Batched pgvector RAG retrieval and tightened citation-style handling in the slide-generation / content-condensation pipeline, improving both latency and citation correctness.

End-to-End Evaluation System (RBS-3):

The automated evaluation system (eval/) was run: ROUGE, BERTScore, schema compliance, custom PPT rules (must_cover, citation consistency, structure), and factual-accuracy metrics from factual_ground_truth.json, with comparative reports across pure LLM / LLM+RAG / LLM+RAG+DeepSearch on two test cases (Transformer NLP and photovoltaic energy). The blind Likert-5 human-assessment track was conducted.

Integration & Reliability (WBS 3.3):

Unified LLM error handling, one-time-ticket SSE authentication, per-task timeout fallback, and a frontend task-stream refactor were completed, materially improving end-to-end reliability.

3. Quantitative Analysis

3.1 Per-Member Hours by WBS Task - Sprint 4

WBS ID	Task	Owner	Hrs	Status	Compl. %
3.3	Unified LLM-call error handling + per-task timeout fallback + bug-fix pass	Ma Xiaolong	6	Completed	95%
3.3/2.4	SSE one-time-ticket authentication + frontend task-stream hook refactor	Ma Xiaolong	4	Completed	95%
2.3	Batched multi-query RAG retrieval (retrieve_many) latency optimization	Ma Xiaolong	3	Completed	100%
—	Sprint review, planning & team coordination	Ma Xiaolong	1	—	—
3.1	Run automated evaluation system (ROUGE/BERTScore/schema/PPT rules/factual accuracy)	Tang Chengyang	9	In Progress	85%
3.2	DeepSearch (Tavily) comparative evaluation runs across the three pathways	Tang Chengyang	5	In Progress	80%
2.2.3	Multi-model horizontal comparison runs (Qwen / GLM / DeepSeek)	Zhang Ruiqi	8	In Progress	85%
3.x	Blind human assessment (Likert-5 peer evaluation)	Zhang Ruiqi	6	In Progress	75%

3.2 Group Total Hours per WBS Section (cumulative, all sprints)

WBS Section	Budget (hrs)	Cumulative Hrs	Spent % of Budget	Task Compl. %	Schedule Variance
1. Project Planning & Design	40	40	100%	100%	On schedule

2. Core Module Development	130	~133	102%	~98%	Complete (final hardening)
3. Integration, Evaluation & Delivery	50	~51	102%	~88%	On track for 06-14
Total	220	~224	102%	~94%	On schedule

Note: this period's reporting team comprises the three members who submitted biweekly reports; one team member did not submit and is therefore excluded from this report. The slight budget overrun on Sections 2 and 3 reflects the final hardening and validation workload of this phase.

4. Risk Analysis & Management

#	Risk	L	I	Mitigation / Action	Status
R1	Multi-model benchmark execution slipping past the validation window	Med	Med	Harness reused; benchmark batched and executed this sprint; model-selection recommendation in consolidation	Closing
R2	Evaluation-phase compression (automated + human-eval both slip)	Low	Med	Evaluation harness reused across all tracks; human-eval rubric finalized early; both tracks run in parallel	Actively managed
R3	External data-source instability (Tavily / DashScope rate limits, latency)	Med	Med	Friendly error mapping + bounded retrieval timeout with graceful degradation; per-batch embedding timeout	Mitigated (by design)
R4	Single-developer concentration of core/integration knowledge (bus factor)	Med	Med	STARTUP.md deployment guide; documented module boundaries; reproducible Dockerized bring-up	Managed
R5	Reduced reporting headcount concentrating delivery workload on the team leader	Med	Med	Core implementation/integration consolidated under the team leader; remaining members on evaluation/testing; scope re-prioritized for 06-14	Actively managed
R6	User trust / acceptance variance of AI-generated content across domains	Low-Med	Med	Citation markers + source-type labelling in exports; stricter citation-style rules; human-eval surfaces domain gaps	Monitoring

Overall risk posture: the realized high-impact technical risk of prior sprints (content-completion deadlock) remains closed, and this sprint further hardened the live generation path. Remaining risks are execution- and delivery-bound rather than design-bound, and are under active management ahead of the June 14 delivery.

5. Upcoming Plan (Final Sprint -> June 14)

Priority	Action	Owner	Target
P0	Consolidate multi-model + RAG/DeepSearch + human-eval results into the technical validation report	Tang Chengyang / Zhang Ruiqi	RBS-3
P0	Final end-to-end regression across full stack; close remaining integration bugs	Ma Xiaolong	WBS 3.3
P1	Finalize DeepSearch query-refinement tuning + complete batch automated-evaluation runs	Tang Chengyang	WBS 3.1/3.2
P1	Deliver model-selection recommendation for production outline generation	Zhang Ruiqi	WBS 2.2.3
P2	Package prototype deployment + finalize technical documentation delivery	Ma Xiaolong	WBS 3.4

Ma Xiaolong (2353814 - Team Leader)

This-sprint hours: 14

Work completed this sprint (reliability hardening of the integrated prototype - WBS 3.3, 2.2-2.4):

- **Unified LLM-call error handling.** Introduced a dedicated LLMClientError and a friendly-error mapper so every upstream LLM failure becomes an actionable, user-facing Chinese message - invalid/expired API key or unauthorized model (401/403), wrong model name or base_url (404), rate-limit/quota exhaustion (429), service unavailable (5xx), timeout, and network errors - while the raw status code and response body are logged but never leaked to the frontend. Applied uniformly to both the blocking and streaming call paths, and surfaced cleanly through the task worker with a per-task hard-timeout fallback.
- **SSE authentication & frontend task-stream refactor.** Hardened the SSE streaming channel: because EventSource cannot send an Authorization header, added a one-time SSE ticket (POST /tasks/{id}/stream-ticket -> short-lived 30s ticket, deleted on use) consumed by GET /tasks/{id}/stream?ticket=...; unified SSE frame construction (event/data/heartbeat/connection comments) and an SSE-timeout error frame; refactored the user frontend onto a dedicated useTaskStream hook (token / progress / done / error handling, reconnection, task resume) and simplified the Chat page.
- **Batched RAG retrieval optimization.** Added a batched retrieve_many path to the vector service that vectorizes all outline-node queries in ceil(N/10) batch embedding calls instead of N single round-trips, cutting multi-chapter content-completion latency and API-call volume (DB queries kept serial since the shared AsyncSession is not concurrency-safe and pgvector lookups are cheap).
- **Citation-style correctness.** Tightened the citation_style rules in the slide-generation prompts (default to summary; forbid the data tag on content containing no figures; concrete snippet examples) and added a citation-style normalization pass in the worker, reducing mislabeled and over-claimed citations in generated slides.
- **Bug-fix pass & coordination.** Diagnosed and fixed assorted defects across the LLM client, vector service, task worker, task/session controllers and the frontend; maintained STARTUP.md; led sprint review, planning and team coordination.

Status / Next steps: The live generation path is materially more robust. Continue final end-to-end regression, close remaining edge-case bugs, and package the technical-documentation deliverable for June 14.

Tang Chengyang (2350434)

This-sprint hours: 14

Work completed (mapped to WBS 3.1, 3.2 - evaluation & testing):

- **Automated evaluation system.** An automated evaluation system (eval/) has been built around the PPT slide-generation stage, covering ROUGE, BERTScore, schema compliance, custom PPT rules (must_cover, citation consistency, structure, etc.), and factual-accuracy metrics based on factual_ground_truth.json.
- **Three-pathway comparison.** The system supports one-click scoring and the generation of comparative reports for three pathways: pure LLM, LLM+RAG, and LLM+RAG+DeepSearch (deep_search_enabled / Tavily in the codebase).
- **Test cases & reporting.** The system now includes completed evaluation scripts, two test cases (Transformer NLP and photovoltaic energy), and a template for generating comparative reports under bundled examples.
- **Findings.** Preliminary results validate RAG's effectiveness in grounding precise facts from the knowledge base, and DeepSearch's added value in supplementing web-based and time-sensitive claims.

Status / Next steps: Finalize DeepSearch query-refinement tuning, complete the batch automated-evaluation runs, and consolidate the results into the technical validation report.

Zhang Ruiqi (2352017)

This-sprint hours: 14

Work completed (mapped to WBS 2.2.3, 3.x - evaluation & testing):

- **Multi-model horizontal comparison.** Conducted a horizontal comparison of several mainstream large language models - such as Qwen, GLM, and DeepSeek - on the outline-generation task. This comprehensive analysis covers generation stability, output quality, and response efficiency, providing data-driven support for the final technology selection.
- **Human assessment.** Conducted a small-scale peer (blind) evaluation to obtain subjective scores on the generated outlines and content, focusing on clarity, completeness, and presentation readiness from a typical user's perspective.

Status / Next steps: Deliver the model-selection recommendation for production outline generation, and fold the human-assessment scores into the consolidated technical validation report.